AIDCF

AI Data Centre Facilities Assurance Framework

Capability-centred, evidence-gated and graph-native assurance for AI infrastructure

Figure 1. AIDCF canonical assurance argument.

Field	Value
Artifact	AIDCF-WP-001
Version	3.0
Status	Validation Candidate for Independent Review
Author	Siva Sethumadhavan
Canonical baseline	AIDCF-007A: 15 domains, 90 families and 360 controls
Implementation status	Synthetic, non-production reference implementation
Publication boundary	Research framework; not law, certification, accreditation, engineering approval or hazardous-test authorisation

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Abstract

Artificial intelligence workloads create a capability system that couples dense electrical demand, heat rejection, liquid and air cooling, accelerated compute, checkpoint storage, high-bandwidth fabrics, orchestration, control planes, operational teams and external providers. Established facility, technology, cybersecurity and AI-governance approaches address important parts of this landscape, but do not necessarily provide one end-to-end argument from a specific AI service outcome to infrastructure reality, credible failure, implemented safeguards, evidence and decision. AIDCF addresses this perceived integration gap.

The framework defines a Capability Instance as the bounded subject of assurance and traces required Service Outcomes through functions, systems, assets, interfaces, dependencies, fault domains, operating states, failure paths, safeguards, controls, evidence, assessments, critical gates and accountable decisions. Its approved baseline contains fifteen domains, ninety control families and 360 controls. The method preserves UNKNOWN, separates intended, designed, installed, commissioned, observed and tested states, and prevents unrelated strengths or aggregate scores from overriding a controlling gate.

Version 3.0 consolidates the AIDCF artifact stack into a coherent research narrative. It documents the development protocol, related-work boundaries, assurance theory, domain derivation, system-of-systems architecture, control and evidence models, assessment and gate logic, a machine-executable publication model and a complete synthetic worked case. It also separates approved baseline content from controlled-review drafts, synthetic demonstrations and proposed extensions. AIDCF remains a proposed research framework requiring independent review, real-world pilots, calibration and multi-site validation.

Executive summary

Problem pattern	AIDCF response	Decision value
Facility and AI-platform reviews are disconnected	One capability chain links outcomes to power, thermal, compute, storage, network, operations and providers.	Exposes cross-domain dependencies and failure propagation.
Design intent is confused with operating reality	States are explicitly separated and evidence is assertion-specific.	Restricts reliance to what was actually observed or tested.
Redundancy masks common mode	Dependencies, routes, controls, people and recovery resources are first-class.	Challenges superficial diversity claims.
Missing evidence is forced into a score	UNKNOWN is visible and non-favourable.	Prevents false certainty.
Aggregate scores overwhelm critical failure	Critical gates are non-compensating.	Protects decision prerequisites.
Automation silently exercises authority	Tools validate and recommend; accountable people decide.	Preserves governance and professional responsibility.

The paper does not claim that no adjacent method exists. It claims that the compared approaches leave a practical integration problem when a decision requires simultaneous reasoning across facility engineering, AI technology, operations, evidence and accountable assurance. This claim is a research proposition to be tested, not a settled conclusion.

Publication posture
Public review draft. Suitable for expert critique, pilot recruitment and research timestamping. Not suitable for certification, accreditation, universal conformance claims or real-facility assurance.

1. Formal research problem

The research problem is to determine whether a repeatable assurance method can reconstruct a bounded conclusion about a defined AI infrastructure capability from: service outcomes; architecture and dependencies; credible adverse conditions; effective safeguards; sufficient evidence; non-compensating prerequisites; competent review; and accountable decision authority.

Research question	Proposition	Empirical test
Can assessors define the same bounded subject?	Capability-first scoping reduces ambiguity.	Independent scoping and inter-rater agreement.
Can facility and AI-platform layers be traced end to end?	Canonical objects and relationships enable reconstruction.	Graph completeness and reverse-trace tests.
Can hidden common modes be surfaced?	Dependency and fault-domain analysis improves discovery.	Compare findings against conventional domain reviews.
Can score gaming be constrained?	UNKNOWN and gates resist favourable denominator manipulation.	Adversarial fixtures and re-performance.
Can decisions remain explainable after change?	Versioned identities and event impact support supersession.	Change-event and affected-decision tests.

2. Claimed contribution and novelty

AIDCF does not claim novelty in power engineering, thermal engineering, systems reliability, cybersecurity, AI risk management, evidence management or assurance individually. Its claimed contribution is the disciplined integration of those elements around a bounded AI Capability Instance, with explicit graph traceability and decision constraints.

Contribution	Distinctive element	Current evidence status
Capability-centred assurance	Service Outcomes precede components and controls.	Conceptual and synthetic demonstration.
Cross-domain failure reasoning	Facility, technology, people and providers share one dependency graph.	Scenario and architecture artifacts; real pilots pending.
Evidence-gated conclusion	Evidence quality and UNKNOWN are first-class	Synthetic evidence fixtures.

Contribution	Distinctive element	Current evidence status
	objects.	
Gate-dominant decision	Critical prerequisites remain separate from scores.	Synthetic gate records and anti-gaming test.
Machine-executable lineage	Stable IDs, JSON Schema, JSON-LD, RDF/SHACL and manifests.	Reference package; independent runtime review pending.

3. Framework-development protocol

The protocol below makes the construction process explicit. It is a retrospective methodological specification for future reproducible maintenance. A public source log, coding register and reviewer record remain required before the derivation can be considered independently reproducible.

Stage	Procedure	Required record
Corpus definition	List included standards, methods, reference architectures and AIDCF source artifacts by edition and scope.	Source register and inclusion rationale.
Concept extraction	Capture terms, objects, relationships, states, evidence and decision concepts.	Extraction matrix with source reference.
Semantic normalization	Separate overloaded concepts such as hazard, cause, failure mode, effect, consequence, safeguard and control.	Terminology decision log.
Domain clustering	Group concepts using coverage, separability and traceability tests.	Cluster matrix and rejected alternatives.
Control synthesis	Express outcome-linked, assessable requirements with evidence and failure mappings.	Control design record and duplicate review.
Assurance design	Define assessment states, UNKNOWN, gates, scores, decisions and reporting.	Rule register and test cases.
Machine formalisation	Assign identifiers, schemas, graph shapes and conformance fixtures.	Release manifest and validation results.
Challenge and approval	Multidisciplinary peer review, disagreement resolution and independent challenge.	Reviewer register, comments and dispositions.

4. Related work and comparative gap

The comparison is directional and limited to cited public scope statements. It is not a clause-by-clause equivalence analysis. ISO/IEC 22237-1 describes general principles, common aspects, facilities and infrastructure classification, and operation and management references, while excluding selection and configuration of IT and network equipment [1]. The Uptime Tier Standard provides an objective basis for comparing site-infrastructure topology and differentiates four classifications [4]. ANSI/TIA-942-C specifies data-centre and computer-room infrastructure requirements and includes telecommunications, power, cooling, architecture, fire protection, safety and physical security [5]. NIST AI RMF 1.0 provides a voluntary AI risk-management framework organised around Govern, Map, Measure and Manage [2] [3].

Approach	Primary public scope	Integration gap AIDCF addresses	No equivalence claim
ISO/IEC 22237	Data-centre facility and infrastructure concepts, classes and management references.	Connection to configured AI compute, storage, fabrics and service outcomes.	AIDCF does not claim ISO conformity.
Uptime Tier topology	Site-infrastructure functionality, capacity and expected availability by topology.	Capability-specific workload, data, fabric, control-plane and evidence reasoning.	Tier evidence is bounded design evidence.
ANSI/TIA-942-C	Physical infrastructure requirements across multiple disciplines.	One assurance argument from AI outcome to operational evidence and decision.	AIDCF does not certify TIA-942.

Approach	Primary public scope	Integration gap AIDCF addresses	No equivalence claim
NIST AI RMF	Voluntary management of AI risk and trustworthiness.	Physical and infrastructure assurance below the AI system.	AIDCF is complementary, not a profile of NIST AI RMF.
AITG	Trust, authority and exposure reasoning.	Infrastructure reality and physical failure paths.	Federate decisions; do not merge methods.

5. Assurance theory

AIDCF defines assurance as bounded confidence in a proposition about a specific capability, for identified users and intended use, within a declared boundary, operating states and period. Assurance is not the inventory of controls, a maturity score or a topology label.

Figure 2. Forward assurance construction and reverse reconstruction.

Proposition layer	Question
Outcome	What must the capability achieve, within what tolerance?
Architecture	Which systems, interfaces and resources implement that result?
Dependency	What internal, shared, external or recovery-only reliance exists?
Failure	How can the outcome be degraded or lost?
Safeguard	What prevents, detects, contains, recovers or adapts?
Evidence	What supports or challenges each assertion?
Gate	Which prerequisite cannot be compensated?
Decision	What reliance is permitted, for whom and for how long?

Conceptual expression

Bounded Assurance = f(Outcome Definition, Architecture and Dependency Confidence, Safeguard Effectiveness, Evidence Confidence, Gate State, Review Quality, Decision Authority). This is a dependency expression, not a numeric multiplication formula.

6. Canonical terminology and acronyms

Term	Canonical meaning
Capability Instance	A specific deployment bounded by owner, location, architecture, demand, criticality, service requirements and obligations.
Service Outcome	A measurable availability, performance, integrity, recovery or restraint condition.
Critical Function	A function whose degradation materially affects a Service Outcome.
Dependency	Reliance on an entity, resource, condition, provider, person or decision.
Fault Domain	Objects expected to be affected by the same failure.
Failure Scenario	Initiating condition through propagation, consequence, safeguards and recovery.
Evidence Object	Controlled item supporting, challenging or qualifying an assertion.
Critical Gate	Non-compensating prerequisite constraining an Assurance Decision.
Assurance Decision	Bounded conclusion issued by accountable authority.
UNKNOWN	State where sufficient and reliable determination cannot be made.

Acronym	Meaning
AIDCF	AI Data Centre Facilities Assurance Framework

Acronym	Meaning
AITG	AI Trust Graph
APS	AIDCF Assurance Principle
MCF	Master Control Framework identifier
CDU	Cooling Distribution Unit
SHACL	Shapes Constraint Language
RDF	Resource Description Framework
TEVV	Testing, evaluation, verification and validation

7. Status taxonomy and component maturity

Status	Validation Candidate for Independent Review	Use in this whitepaper
Approved baseline	Constitutionally approved AIDCF identity and cardinality baseline.	AIDCF-007A 15/90/360 only.
Controlled-review draft	Substantive methodology awaiting multidisciplinary review.	Most AIDCF standards and handbooks.
Synthetic demonstration	Non-production data and execution for method testing.	AIDCF-015 v1.2.
Proposed research extension	Concept requiring governance and validation before operational use.	Maturity model, claim patterns and future-state ideas.
Externally validated	Independent execution or empirical validation completed.	Not yet achieved.

This taxonomy prevents the whitepaper from presenting every component as equally mature. Canonical identity can be approved while substantive control adequacy, gate calibration and real-world effectiveness remain open for review.

8. Why fifteen domains

The fifteen-domain structure applies three design tests. Coverage asks whether every material condition can be anchored without arbitrary omission. Separability asks whether a group has coherent ownership, competence and evidence. Traceability asks whether cross-domain effects can remain explicit rather than hidden inside broad categories.

Cluster	Domains	Rationale
Direction and subject	GOV, CAP	Separate governance and decision rights from the capability being assured.
Dependency and place	DEP, PHY	Make hidden reliance and physical boundary explicit.
Facility engineering	POW, THM, ENV	Different mechanisms, expertise, telemetry and failure modes.
AI technology chain	CMP, STO, NET	Distinct but tightly coupled compute, data and communication functions.
Operation and recovery	OPS, RES	Separate routine control from continuity, crisis and restoration.
Ecosystem and authority	SUP, SEC	Separate supplier/lifecycle reliance from digital and OT authority.
Assurance integrity	EVA	Independently govern evidence, assessment, gates and decision integrity.

This remains a design hypothesis. Completeness must be tested through expert review, real incidents, pilot assessments and analysis of unmapped requirements.

9. Domain model and control baseline

Code	Domain	Primary gate mapping
GOV	Governance, Accountability and Assurance Direction	CG-13, CG-14
CAP	AI Capability, Service Outcome and Demand Management	CG-01, CG-02
DEP	Dependency, Concentration and External Reliance	CG-03, CG-12
PHY	Physical Site, Building and Security Infrastructure	CG-02, CG-04
POW	Electrical Power Infrastructure and Energy Continuity	CG-05
THM	Cooling, Thermal and Fluid Infrastructure	CG-06
ENV	Environmental and Resource Performance	CG-02, CG-04
CMP	AI Compute Infrastructure	CG-08
STO	AI Storage and Data Infrastructure	CG-08, CG-10
NET	AI Network and Interconnect Architecture	CG-08
OPS	Building Operations and Service Management	CG-07, CG-13, CG-14
RES	Resilience, Continuity and Recovery	CG-09, CG-10
SUP	Supply Chain, OEM and Ecosystem Assurance	CG-03, CG-13
SEC	Digital, OT and Control-Plane Security	CG-07, CG-15
EVA	Evidence, Assessment and Assurance Management	CG-11, CG-12

Each domain contains six families and twenty-four controls, yielding ninety families and 360 controls. Stable identity does not imply immutable wording. Meaning-changing replacements require new identifiers and explicit supersession.

10. System-of-systems reference architecture

Figure 3. Reference architecture with assurance and resilience rails.

The vertical service stack traces how external resources and facilities enable AI technology and AI services to achieve business outcomes. The assurance rail attaches controls, evidence, assessments, gates and decisions to every layer. The resilience rail attaches states, dependencies, fault domains, failure paths, safeguards and recovery.

Mandatory view	Required content
Capability	Owner, lifecycle, outcomes, tolerances and criticality.
Boundary	Physical, logical, organisational, provider and temporal limits.
Service chain	Functions and interfaces from workload to facility resources.
Dependency	Internal, shared, external and recovery-only reliance.
State and capacity	Normal, maintenance, degraded, emergency, recovery and transition states; installed, available, resilient and committed capacity.
Failure and safeguard	Initiating condition, propagation, consequence, safeguards, residual path and recovery.
Assurance attachment	Controls, evidence, assessments, findings, gates, score and decision.

11. Capability service chains

Capability pattern	Required functions	Representative adverse conditions
Distributed training	Scheduling, accelerator capacity, collective communication, training data, checkpointing,	Fabric partition, thermal throttling, checkpoint

Capability pattern	Required functions	Representative adverse conditions
	power and cooling.	loss, scheduler failure.
Inference	Request routing, model serving, accelerator availability, cache/data access and latency observability.	Tail-latency excursion, overload, model-store or network degradation.
Fine-tuning	Data preparation, workload placement, artifact management and reproducible recovery.	Configuration drift, dataset access loss, interrupted checkpoint cycle.
Sovereign AI	Boundary enforcement, local operations, supply visibility, restricted administration and continuity.	Remote authority, jurisdictional dependency, external telemetry or support concentration.
Shared AI platform	Tenant isolation, fair scheduling, capacity reservation, telemetry and service management.	Noisy neighbour, privilege misuse and resource contention.

12. Dependency, capacity and fault-domain analysis

Capacity state	Meaning
Installed	Nominal capacity physically present.
Available	Usable after condition, maintenance and derating.
Resilient	Available in a defined adverse condition.
Committed	Allocated or contractually reserved.
Headroom	Remaining usable capacity beyond demand and commitments.

Apparent diversity is not independence. Two paths may share utility supply, fuel, heat rejection, management network, firmware, time service, carrier, physical route, operator or OEM. A dependency assertion records consumer, provider, direction, condition, required capacity, criticality, evidence, confidence, owner and validation date.

13. Threat, hazard and failure model

Object	Analytical question
Threat source	Who or what could intentionally create an adverse event?
Hazard	What condition has potential to cause degradation or harm?
Cause	Why did the failure mode occur?
Failure mode	How did the required function fail?
Local effect	What occurred at the failed object?
Propagation	Which dependencies transmitted the effect?
Consequence	Which Service Outcome or stakeholder interest was affected?
Safeguard	What anticipates, prevents, detects, contains, recovers or adapts?
Residual path	What material route remains after safeguards?

AIDCF can consume FMEA/FMECA, fault-tree, event-tree, bow-tie, HAZOP-style and graph-traversal outputs. The framework does not replace competent use of those techniques or applicable engineering and safety requirements.

14. AI-native failure scenarios

Scenario	Failure path	Primary domains
Accelerator thermal throttling	Liquid-flow degradation → temperature rise → frequency reduction → collective desynchronisation → checkpoint exposure	THM, CMP, NET, OPS, RES

Scenario	Failure path	Primary domains
Fabric congestion	Oversubscription or path imbalance → collective latency → stalled or elongated job	NET, CMP, DEP, OPS
Checkpoint corruption	Metadata or storage degradation → apparently successful but unusable recovery point	STO, NET, RES, EVA
Scheduler collapse	Stale capacity or policy failure → placement failure despite installed accelerators	CMP, CAP, OPS, SEC
Inference instability	Load spike, cache miss, fabric or thermal degradation → tail-latency breach	CAP, CMP, NET, STO, THM
Sovereign-boundary failure	Remote support or telemetry path → declared authority or jurisdiction boundary breach	GOV, DEP, SUP, SEC
AI capacity cliff	Committed demand plus degraded resilient capacity → insufficient usable headroom	CAP, POW, THM, CMP, NET

15. Safeguards and resilience

Safeguard role	Examples	Evidence focus
Anticipate	Demand forecasting, trend and concentration review.	Model validity and review.
Prevent	Redundancy, authority enforcement and environmental control.	Design and implementation.
Detect	Telemetry, alarms, anomaly detection and integrity checks.	Coverage, source health and latency.
Contain	Fault isolation, segmentation, safe load reduction and leak isolation.	Activation and effectiveness.
Recover	Restore configuration, capacity, data and service.	Sequencing, elapsed time and proof of outcome.
Adapt	Architecture, procedure and operating-envelope change.	Approved change and supersession.

A standby mechanism is not an effective safeguard if it depends on the failed controller, lacks capacity, cannot activate in time or creates an unsafe restoration path.

16. Evidence model

Dimension	Assessment question
Provenance	Can source and lineage be identified?
Freshness	Does evidence represent the relevant state and period?
Coverage	Does it cover the required population, path, state and time?
Reliability	Was it produced by a controlled and repeatable method?
Traceability	Can it be linked to assertion, finding, gate and decision?
Integrity	Is inappropriate alteration detectable and custody controlled?
Sufficiency	Is the evidence set adequate for the decision and assurance depth?

Evidence is assertion-specific. A signed design supports intent but not necessarily implementation. Telemetry can support current operation but not historical commissioning. A test can support one state or scenario without proving all conditions. AI-assisted extraction or summarisation creates candidate evidence only; it does not independently corroborate its source.

17. Assessment methodology

Stage	Key output
Initiate	Decision question, sponsor, users and charter.
Profile	Capability, Service Outcomes, architecture and operating envelope.
Scope	Boundary, dependencies, states, criteria and exclusions.
Tailor	Applicable controls, gates, evidence and assurance depth.
Plan	Methods, sampling, tests, safety and review.
Collect	Controlled Evidence Objects and restrictions.
Validate	Provenance, integrity, freshness, coverage and sufficiency.
Assess	Control, safeguard, scenario and domain results.
Analyse	Propagation, common mode, concentration and residual paths.
Gate	Applicable CG-01 through CG-15.
Find	Findings, UNKNOWN, limitations and recommendations.
Review	Technical and independent challenge.
Decide	Bounded Assurance Decision.
Report	Executive, technical and machine-readable packages.
Reassess	Change, incident, expiry and supersession.

18. Critical gates and governance

Gate identity and domain mappings are canonical in the registry. The explanatory names and decision effects below are public-review profiles, not replacements for the controlled gate registry. They must be approved through AIDCF governance before being called canonical gate definitions.

Gate	Proposed explanatory profile
CG-01	Capability definition
CG-02	Boundary and operating envelope
CG-03	Dependency and provider confidence
CG-04	Physical and environmental suitability
CG-05	Electrical continuity
CG-06	Thermal and fluid continuity
CG-07	Operational and control authority
CG-08	AI technology service-chain integrity
CG-09	Containment and resilience
CG-10	Recovery and restoration
CG-11	Evidence integrity and sufficiency
CG-12	Traceability and completeness
CG-13	Governance, competence and accountability
CG-14	Validity, change and continuing assurance
CG-15	Digital and OT security authority

Controlled states are PASS, FAIL, UNKNOWN, APPROVED_NA and NOT_ASSESSED. Conditions are separate fields. FAIL demonstrates that a prerequisite is not met. UNKNOWN means the evidence cannot establish the state. Neither can be treated as favourable.

19. Scoring and decision model

Scores support synthesis and trend analysis. They do not create assurance, neutralise UNKNOWN, override a gate or replace accountable judgement.

Decision state	Meaning
REASONABLY ASSURED	Required evidence and gates support the bounded claim. This is an AIDCF internal state, not a regulated assurance level.
CONDITIONALLY ASSURED	The core claim is supported subject to named, time-bound conditions and no controlling failure.
LIMITED	Material scope, evidence or UNKNOWN restricts reliance.
NOT ASSURED	Evidence demonstrates material failure or a controlling gate fails.
UNKNOWN / INCONCLUSIVE	Evidence, method or boundary cannot support determination.
SUSPENDED	A prior decision is invalidated by change, incident, expiry or gate state.
WITHDRAWN	Decision removed after material integrity or governance failure.
SUPERSEDED	New approved snapshot replaces the earlier decision.

20. Machine-executable publication and conformance

Component	Purpose	Proof boundary
CSV registries	Canonical identities and vocabularies.	Identity and completeness.
JSON Schema	Structural validation.	Shape, not factual truth.
JSON-LD	Linked-data exchange.	Context and identity.
RDF/OWL	Semantic classes and relationships.	Declared semantics.
SHACL	Graph-shape and cardinality constraints.	Graph conformance.
Fixtures	Positive, negative, migration and anti-gaming tests.	Expected rule behaviour.
Manifest and hashes	Release inventory and integrity.	File-level reproducibility.
Conformance report	Captured execution, versions and limitations.	Technical result only.

Machine conformance cannot establish engineering adequacy, safety, operating effectiveness or assurance over a real facility. AIDCF-015 v1.2 is a canonical-baseline-aligned synthetic reference implementation, subject to independent T7 review and stated runtime limitations.

21. Complete synthetic worked case

Figure 4. Synthetic service chain and assessed boundary.

The case demonstrates method execution without representing a real organisation or facility. The Capability Instance is a synthetic liquid-cooled distributed-training service. Required outcomes are job completion, checkpoint integrity, controlled thermal operation and bounded restoration. The scope contains one cluster service chain and twelve representative control assessments. This equals 3.3333% of the 360-control catalogue and cannot support full-coverage claims.

Case-local ID	Canonical domain	Assertion	Evidence object
SYN-CTL-01	CAP	Outcomes and operating envelope are defined.	EV-001 capability and outcome record.
SYN-CTL-02	DEP	Material power, cooling, network and storage dependencies are mapped.	EV-002 dependency map.
SYN-CTL-03	POW	Available and resilient capacity are distinguished.	EV-003 capacity study.
SYN-CTL-04	THM	CDU and cooling-loop capacity	EV-004 thermal model.

Case-local ID	Canonical domain	Assertion	Evidence object
		supports declared load.	
SYN-CTL-05	CMP	Cluster topology and scheduler state are controlled.	EV-005 compute baseline.
SYN-CTL-06	STO	Checkpoint integrity and restoration are addressed.	EV-006 storage record.
SYN-CTL-07	NET	Fabric topology and degraded-path performance are addressed.	EV-007 network record.
SYN-CTL-08	OPS	Operational authority and response procedures are assigned.	EV-008 procedure record.
SYN-CTL-09	RES	Containment and recovery dependencies are analysed.	EV-009 scenario record.
SYN-CTL-10	SEC	Privileged control-plane authority is controlled.	EV-010 access record.
SYN-CTL-11	EVA	Evidence lineage and integrity values are retained.	EV-011 evidence manifest.
SYN-CTL-12	EVA	Decision, limitations and validity are recorded.	EV-012 decision record.

22. Worked-case assessment results

Case-local ID	Procedure	Result	Decision effect
SYN-CTL-01	Inspect capability/outcome record and trace architecture linkage.	CONFORMING	Supports CG-01.
SYN-CTL-02	Trace upstream, shared and recovery dependencies.	PARTIAL	CG-03 remains constrained.
SYN-CTL-03	Reconcile installed, available, resilient and committed capacity.	CONFORMING	Supports CG-05 in synthetic scope.
SYN-CTL-04	Review heat-load model and CDU dependency.	CONFORMING	Supports CG-06 in synthetic scope.
SYN-CTL-05	Inspect topology, scheduler and health data.	CONFORMING	Supports CG-08.
SYN-CTL-06	Inspect checkpoint evidence and restoration result.	UNKNOWN	CG-10 remains UNKNOWN.
SYN-CTL-07	Review fabric topology and degraded-path test.	PARTIAL	Limits CG-08 confidence.
SYN-CTL-08	Review procedures and authority.	CONFORMING	Supports CG-07.
SYN-CTL-09	Review containment and recovery scenario.	UNKNOWN	CG-09 and CG-10 constrained.
SYN-CTL-10	Review privilege and change evidence.	CONFORMING	Supports CG-15.
SYN-CTL-11	Reperform hashes and lineage.	CONFORMING	Supports CG-11.
SYN-CTL-12	Reperform decision and limitation wording.	CONFORMING	Supports decision integrity.

Gate outcome: no real-facility gate conclusion is made. Within the synthetic case, recovery and containment remain materially UNKNOWN. Coverage is 3.3333%. The appropriate bounded decision is LIMITED. The case prohibits any claim about a real facility, complete control coverage, certification or production conformance.

Worked-case conclusion
LIMITED, synthetic, non-production and non-certification. The favourable results for selected controls do not overcome low catalogue coverage or material UNKNOWN.

23. Reporting and permitted descriptive claims

Claim pattern	Minimum disclosure	Must not imply
AIDCF-aligned	Artifact versions, adopted concepts and deviations.	Complete assessment.
Assessed using AIDCF	Capability, profile, scope, states, period, evidence access, gates and issuer.	Certification or universal safety.
AIDCF assurance-ready	Prepared architecture, evidence ownership and programme governance.	That controls are effective.
Continuing-assurance enabled	Approved monitoring, evidence-health and reassessment triggers.	Perpetual validity or autonomous certification.

These are proposed descriptive claim patterns, not badges or conformance marks. Public mark use, eligibility, evidence packages, withdrawal and challenge procedures require separate governance. “AIDCF Certified” and “AIDCF Accredited” remain prohibited.

24. AIDCF and AI Trust Graph

Integration pattern	Use	Constraint
Reference	One method points to an entity owned by the other.	Do not duplicate authoritative properties.
Mapping	Classes or relationships are broader, narrower, close or related.	Record mapping type and reviewer.
Shared identity	The same real-world instance appears in both.	One global identity with method-specific views.
Boundary handoff	An infrastructure service becomes an AI-system dependency or enforcement point.	Record conditions and effective period.
Scenario composition	Infrastructure failure path connects to trust or authority path.	Retain native semantics.
Evidence reuse	One Evidence Object supports distinct assertions.	Assess relevance separately.
Decision federation	Separate conclusions appear in one package.	No automatic best- or worst-case aggregation.

AIDCF represents infrastructure reality and physical/digital failure. AITG represents trust, authority and exposure. The integration standard is a controlled-review draft and product architecture, proprietary algorithms, deployment design and confidential operational data remain outside its public scope.

25. Adoption, maturity and economics

The maturity and economic models are proposed research extensions. They are included for pilot design and are not approved AIDCF baseline content.

Maturity stage	Observable capability	Caution
1 Foundational	Named owner, bounded capability and critical evidence inventory.	Not an assurance conclusion.
2 Managed	Repeatable ownership, architecture and issue management.	Local consistency only.
3 Defined	Standard profiles, canonical IDs and traceable assessments.	Requires quality review.
4 Measured	Evidence health, scenario tests and calibrated indicators.	Requires comparable scope and method.
5 Adaptive and continuously evidenced	Independent challenge, timely suspension and controlled evolution.	Must not be branded “Assured.”

Economic mechanism	Pilot measure
Stranded accelerator capacity	Unavailable accelerator-hours attributable to facility or service-chain

Economic mechanism	Pilot measure
	constraints.
Training interruption	Rework, restart frequency and lost job time.
Inference degradation	Service-outcome breach frequency and duration.
Emergency intervention	Unplanned specialist, vendor and logistics cost.
Delayed programme value	Decision delay attributable to infrastructure uncertainty.
Evidence friction	Repeated evidence collection and review effort.

26. Validation status and roadmap

Figure 5. Validation roadmap from synthetic execution to independent evidence.

Validation item	Current status	Required closure
Canonical identity and cardinality	Approved 15/90/360 baseline.	Maintain controlled registry and supersession.
Synthetic package	Baseline-aligned, twelve representative controls.	Retain limited, synthetic claim.
Independent T7 execution	Pending.	Independent rerun and signed report.
RDF/SHACL runtime	Attempted when runtime available; static tests retained.	Capture tool versions and executed output.
Real capability pilot	Not completed.	Authorised multidisciplinary pilot.
Inter-rater reproducibility	Not completed.	Independent teams and disagreement analysis.
Multi-site calibration	Not completed.	Different architectures, providers and operating models.
Public benchmark corpus	Not completed.	Anonymised permitted cases and expected outputs.

27. Threats to validity

Threat	Effect	Mitigation
Author concentration	Framework design and review may reflect one viewpoint.	Independent multidisciplinary reviewers and public dispositions.
Source-selection bias	Compared literature may omit adjacent methods.	Publish source register, search terms and inclusion criteria.
Construct validity	Domains, gates or maturity stages may not measure intended constructs.	Expert review, pilot observation and revision.
Synthetic-case bias	Fixtures can reflect designed rules rather than operational complexity.	Real evidence, messy data and external cases.
Inter-rater variability	Different assessors may interpret requirements differently.	Calibration, decision rules and blinded comparison.
Commercial confirmation bias	Economic value may be overstated.	Predefined pilot measures and transparent null results.
Security disclosure risk	Detailed dependency paths may expose sensitive infrastructure.	Classification, minimisation and controlled reporting.

28. Limitations and exclusions

- AIDCF is a research framework and controlled methodology, not an accredited standard or certification scheme.

- The control baseline has approved identity and cardinality, but substantive adequacy still requires multidisciplinary review and pilot evidence.
- The synthetic implementation exercises twelve controls, not the full 360-control universe.
- No statistically meaningful real-facility or multi-site validation has been completed.
- Proposed gate profiles, maturity stages, claim patterns and economic measures require governance and calibration.
- Machine conformance does not prove factual truth, engineering adequacy, safety or operating effectiveness.
- The framework does not authorise switching, intrusion, failover, discharge, thermal stress or destructive testing.
- Applicable law, engineering codes, safety rules, contractual duties and professional standards remain authoritative.

29. Conclusion

AIDCF proposes a capability-centred assurance architecture for AI data centres and AI infrastructure. Its contribution is the explicit trace from Service Outcome to architecture, dependency and fault domain; from credible failure to safeguard; from assertion to Evidence Object and assessment; and from critical gate to accountable, bounded decision.

Version 3.0 corrects the narrative and governance weaknesses of the earlier expanded draft. It distinguishes approved baseline content from controlled-review drafts and proposed extensions, qualifies its novelty claim, documents a reproducible maintenance protocol, clarifies gate authority, provides a complete synthetic case and states threats to validity. It does not convert the framework into an established standard. The next source of credibility must be independent execution, multidisciplinary review and empirical pilot evidence.

Final proposition
AIDCF should earn authority through transparent evidence, reproducible method and disciplined claim boundaries, consistent with the assurance doctrine it applies to AI infrastructure.

Appendix A. External-framework crosswalk

Framework	Published scope used here	Typical AIDCF interface	Reference
ISO/IEC 22237-1:2021	General principles, common aspects, facilities/infrastructure classification and management references.	PHY, POW, THM, ENV, OPS and RES evidence.	[1]
NIST AI RMF 1.0	Voluntary AI risk management and trustworthiness; Govern, Map, Measure, Manage.	GOV, CAP, SEC and EVA linkage.	[2][3]
Uptime Tier Standard: Topology	Site-infrastructure topology, capacity components, distribution paths and performance confirmation.	POW, THM, PHY and OPS design evidence.	[4]
ANSI/TIA-942-C	Data-centre infrastructure requirements including telecommunications, power, cooling, architecture, fire protection, safety and physical security.	PHY, POW, THM and NET evidence.	[5]
W3C PROV-O	Provenance representation.	Evidence lineage and transformation.	[6]

This crosswalk is directional and non-normative. It does not establish equivalence, coverage or conformity.

Appendix B. Source and artifact register

ID	Artifact	Status used in v2.0
AIDCF-005	AI Trust Graph Integration Standard v1.0	Controlled-review draft.
AIDCF-007A	Constitutional Amendment and Canonical Registry v1.0	Approved identity/cardinality baseline.
AIDCF-009	Enhanced Practitioner Handbook v1.1	Controlled-review draft.
AIDCF-012	Reporting and Assurance Statement Standard v1.0	Controlled-review draft.
AIDCF-013	AI Data Centre Assurance Reference Architecture v1.0	Controlled-review draft.
AIDCF-014	End-to-End Traceability and Machine-Executable Data Standard v1.0	Controlled-review draft.
AIDCF-015 v1.2	Canonical Baseline Migration Package	Synthetic, non-production implementation; independent T7 pending.

Appendix C. Review and release checklist

Gate	Release requirement	v2.0 status
R1	Narrative follows research sequence and has one conclusion.	Complete.
R2	Novelty claim is qualified.	Complete.
R3	Development protocol, source register and threats to validity are visible.	Protocol complete; detailed extraction register pending.
R4	Gate authority distinguishes canonical and explanatory content.	Complete.
R5	Complete synthetic case and claim boundaries included.	Complete.
R6	External claims have numbered citations and reference details.	Complete for cited public scope statements.
R7	Independent multidisciplinary technical review completed.	Pending.
R8	Independent T7 and semantic runtime evidence complete.	Pending.
R9	Real capability pilot and inter-rater study complete.	Pending.
R10	Licence, DOI and public repository metadata approved.	Pending publication decision.

Appendix D. Authorship, conflicts and contribution statement

Author: Siva Sethumadhavan. Contributions: conceptualisation, methodology design, artifact architecture, control-framework design, writing, synthetic implementation direction and visualisation. Independent technical review has not yet been completed. No claim of external organisational endorsement is made in this document. Any employment affiliation should be added only after applicable review and authorisation.

Conflict-of-interest statement: the author is the originator of AIDCF and therefore has an inherent interest in its adoption. This is mitigated through explicit limitations, open review, publication of negative tests, separation of baseline identity from substantive validation, and a roadmap for independent challenge.

Appendix E. Version history and publication metadata

Version	3.0	Principal change
1.0	Internal draft	Initial consolidated whitepaper.
1.1	Expanded internal draft	Added research, gate, maturity, economics, AITG and validation sections, but retained duplicated narrative.
2.0	Public Review Draft	Rebuilt narrative, qualified novelty, added protocol, status taxonomy, worked case, threats to validity, citations and release gates.

Metadata field	Value
Resource type	Technical report / preprint
Title	AIDCF: AI Data Centre Facilities Assurance Framework
Version	3.0
Creator	Siva Sethumadhavan
Licence	To be selected before public deposit
DOI	To be reserved before final PDF if displayed in document
Related outputs	Canonical registry dataset and synthetic implementation package
Keywords	AI data centre; AI factory; infrastructure assurance; accelerated compute; liquid cooling; evidence; resilience; knowledge graph; critical gates

30. v3.0 release posture and evidence boundary

Version 3.0 is a validation candidate, not a final validated standard. It consolidates the technical, editorial and governance recommendations raised during the v1.1 and v2.0 adversarial reviews. It includes author dispositions, a candidate critical-gate approval package, a derivation-record structure, an independent-review protocol and an authorised-pilot package. It does not claim that independent reviewer acceptance, canonical gate ratification or a real authorised pilot has occurred where no signed evidence is available.

Non-fabrication boundary
No reviewer name, approval signature, pilot sponsor, site, result or authorisation is asserted without a signed source record. The registers below distinguish author-resolved items from evidence still pending.

Component	v3.0 state	Permitted interpretation
Narrative and technical corrections	Author-resolved	Whitepaper text incorporates the identified corrections.
Reviewer dispositions	Author disposition recorded; independent acceptance pending	Issues have proposed resolutions, not third-party endorsement.
Critical-gate registry	Candidate canonical package	Ready for governance review; not independently ratified.
Derivation records	Register structure and current evidence assembled	Partially populated; retrospective source coding remains incomplete.
Pilot package	Authorisation and execution templates complete	No real authorised pilot claimed.
Synthetic implementation	Existing AIDCF-015 v1.2	Technical demonstration only, twelve controls and 3.3333% coverage.

31. Consolidated reviewer-disposition register

The register records the substantive issues raised in adversarial reviews of v1.1 and v2.0. “Author-resolved” means the text or package was changed by the framework author. It does not mean an independent reviewer accepted the disposition.

ID	Issue	Disposition	Status	Residual action
RD-001	Narrative duplicated and concluded too early	Reordered into one research sequence with one conclusion.	Author-resolved	Independent editorial acceptance pending
RD-002	Novelty claim overbroad	Reframed as a perceived integration gap and testable proposition.	Author-resolved	Literature-review challenge pending
RD-003	Derivation not reproducible	Added protocol, source register, extraction and decision-log requirements.	Partially resolved	Retrospective population pending
RD-004	15/90/360 symmetry not technically justified	Separated identity/cardinality design from empirical completeness.	Author-resolved	Domain SME challenge pending
RD-005	Gate profiles not canonical	Created candidate gate-approval package and explicit provisional status.	Partially resolved	Governance ratification pending
RD-006	Scoring not reperformable from whitepaper	Added scoring specification and re-performance requirements.	Partially resolved	Canonical numeric example pending source extraction
RD-007	Architecture too conceptual	Added system-of-systems, authority, evidence and failure viewpoints.	Author-resolved	Engineering review pending
RD-008	Related work too narrow	Expanded source families and comparison criteria.	Author-resolved	Systematic-search completeness pending
RD-009	Worked case insufficiently complete	Added traceability, evidence, result, gate and decision record requirements.	Partially resolved	Canonical MCF mapping must come from executable package
RD-010	Maturity model could be misused	Renamed Level 5 and kept model as research extension.	Author-resolved	Pilot calibration pending
RD-011	Conformance labels may become badges	Reframed as descriptive claim patterns with governance restrictions.	Author-resolved	Legal and mark-governance review pending
RD-012	Economic claims not evidenced	Converted to pilot measures and hypotheses.	Author-resolved	Empirical data pending
RD-013	Security disclosure governance weak	Added information-classification and public-release rules.	Author-resolved	Security review pending
RD-014	Public review not operationalised	Added reviewer instructions, issue taxonomy and disposition workflow.	Author-resolved	Repository/contact metadata pending
RD-015	Independent review absent	Added independent-review board charter and sign-off matrix.	Open	Named independent reviewers pending
RD-016	Real authorised pilot absent	Added pilot charter, authorisation gate and evidence pack.	Open	Sponsor and signed authorisation pending

32. Canonical critical-gate approval candidate

This section defines the candidate governance record required to ratify CG-01 through CG-15. It is proposed for approval and must not be described as independently approved until the approval table is signed by the defined authorities.

Field	Candidate rule
Owner	AIDCF methodology owner.
Approvers	Methodology owner plus independent assurance reviewer and relevant domain specialists.
Approval unit	Gate identity, name, purpose, applicability, evidence floor, state criteria, decision effect and reassessment trigger.
Allowed states	PASS, FAIL, UNKNOWN, APPROVED_NA and NOT_ASSESSED.
Change rule	Meaning-changing revision requires version increment, impact analysis and supersession.
Conflict rule	Conflicting evidence creates UNKNOWN or FAIL according to the approved gate criterion; it cannot be silently reconciled.
Authority boundary	Gate evaluation constrains but does not itself issue an Assurance Decision.

Gate	Candidate name	Minimum evidence floor	FAIL/UNKNOWN trigger	Decision effect
CG-01	Capability definition	Defined Capability Instance, owner, outcomes and criticality	Undefined or materially ambiguous subject	Blocks positive conclusion
CG-02	Boundary and operating envelope	Physical, logical, provider, state, time and demand boundary	Material boundary or tolerance UNKNOWN	Applies ceiling
CG-03	Dependency and provider confidence	Internal, shared, external and recovery dependencies	Hidden common mode or critical provider uncertainty	Constrains reliance
CG-04	Physical and environmental suitability	Site, building, zoning, hazard and environmental evidence	Material suitability deficiency	May produce NOT ASSURED
CG-05	Electrical continuity	Capacity, quality, protection, autonomy and restoration	Controlling path failure or insufficient evidence	Non-compensating
CG-06	Thermal and fluid continuity	Heat-load model, resilient capacity, fluid integrity and recovery	Thermal excursion, common controller or unverified recovery	Non-compensating
CG-07	Operational and control authority	Competence, procedures, alarms, access and safe change	Unclear authority or unsafe control	Blocks operational reliance
CG-08	AI technology service-chain integrity	Compute, storage and network performance and health	Material technology-chain failure	Constrains capability claim
CG-09	Containment and resilience	Fault domains, graceful degradation and containment evidence	Uncontained propagation or untested material scenario	Blocks resilience claim
CG-10	Recovery and restoration	Recovery capacity, sequence, integrity and outcome test	Material recovery path UNKNOWN or failed	Limits or suspends
CG-11	Evidence integrity and sufficiency	Provenance, freshness, coverage, integrity and sufficiency	Evidence cannot support a material assertion	Blocks favourable result
CG-12	Traceability and completeness	Forward/reverse lineage, conflicts, exclusions and UNKNOWN	Orphan result or incomplete chain	Blocks issued statement

Gate	Candidate name	Minimum evidence floor	FAIL/UNKNOWN trigger	Decision effect
CG-13	Governance, competence and accountability	Decision rights, ownership, competence and independence	No accountable authority or material conflict	Blocks issuance
CG-14	Validity, change and continuing assurance	Validity, monitoring, trigger and reassessment records	Material change invalidates conditions	Suspends prior decision
CG-15	Digital and OT security authority	Identity, privilege, segmentation, detection and safe automation	Uncontrolled privileged or automated path	May block operation

Approval status
Candidate canonical package prepared. Formal ratification remains PENDING because no signed multidisciplinary approval record was found or provided.

33. Critical-gate approval record and sign-off matrix

Role	Required review	Decision	Evidence required
Methodology owner	Coherence with AIDCF principles and downstream artifacts	Pending	Signed approval record
Assurance/scoring reviewer	Gate dominance, UNKNOWN and decision effects	Pending	Independent review memo
Electrical SME	CG-05 and cross-domain electrical effects	Pending	Domain disposition
Mechanical/thermal SME	CG-06 and thermal/fluid effects	Pending	Domain disposition
Compute/HPC SME	CG-08 compute service chain	Pending	Domain disposition
Storage/network SME	CG-08 and CG-10 data/fabric effects	Pending	Domain disposition
Operations/resilience SME	CG-07, CG-09, CG-10 and CG-14	Pending	Domain disposition
OT/security SME	CG-07 and CG-15	Pending	Domain disposition
Evidence specialist	CG-11 and CG-12 evidence floors	Pending	Domain disposition
Safety/legal reviewer	Safety and public-claim boundaries	Pending	Review memorandum

A gate becomes CANONICAL only when the required approvals are complete, unresolved objections are recorded and the resulting registry is versioned with an effective date. Author approval alone is insufficient for independent validation.

34. Full derivation-record package

The derivation package is the evidence needed to show how AIDCF was constructed. The table distinguishes records present conceptually from records that still require retrospective population or independent verification.

Record	Required contents	v3.0 state
DR-01 Source register	Title, publisher, edition, scope, access basis, inclusion rationale and extraction date.	Partially populated in references; full register pending.
DR-02 Search protocol	Databases/sites, search strings, dates, inclusion/exclusion criteria and reviewer.	Specified below; execution log pending.
DR-03 Concept extraction matrix	Source passage, concept, class, property, relationship and analyst note.	Template complete; population pending.
DR-04 Terminology log	Ambiguous term, alternatives, adopted meaning and rationale.	Core terminology populated; full decisions pending.
DR-05 Domain cluster matrix	Concept-to-domain mapping, rejected clusters and coverage test.	High-level cluster rationale complete; detailed matrix pending.
DR-06 Control derivation register	Source need, objective, canonical control, duplicate review and evidence mapping.	Canonical registry exists; derivation evidence pending.
DR-07 Gate derivation register	Failure consequence, prerequisite, evidence floor and decision effect.	Candidate package complete; approval pending.
DR-08 Reviewer register	Reviewer competence, independence, comments and dispositions.	Author dispositions complete; independent register pending.
DR-09 Change and supersession log	Changed object, reason, impact, version and migration.	AIDCF-007A and AIDCF-015 migration evidence exists.
DR-10 Validation corpus	Positive, negative, anti-gaming and benchmark cases.	Synthetic fixtures exist; public corpus pending.

34.1 Search and inclusion protocol

- Prioritise official standards bodies, public institutional guidance, peer-reviewed or established assurance literature and authoritative vendor-neutral technical bodies.
- Record exact search terms, date executed, source URL or DOI, edition, access restriction and reviewer.
- Include sources relevant to facilities, AI technology, OT/control systems, reliability, evidence, assurance arguments, provenance and machine validation.
- Exclude unsupported marketing claims, unattributed summaries and sources whose scope cannot be established.
- Treat paywalled normative standards through their accessible abstracts unless licensed full text is available. Do not imply clause-level comparison from public abstracts.

34.2 Comparative analysis criteria

Criterion	Question
Subject	What system or proposition is assessed?
Facility coverage	Are power, thermal, physical and environmental systems included?
AI technology	Are configured compute, storage, fabrics and orchestration included?
Operations	Are operating states, change and recovery addressed?
Failure reasoning	Are common modes and propagation explicit?
Evidence	Are provenance, freshness, coverage and sufficiency explicit?
Uncertainty	Can missing or conflicting evidence remain UNKNOWN?
Decision restraint	Can critical prerequisites override aggregate scores?
Machine traceability	Can conclusions be reconstructed and validated?
Authority	Who may issue, rely on, suspend or withdraw a conclusion?

35. Expanded related-work map

Source family	Explicit public scope used	AIDCF interface	Review caveat
ISO/IEC 22237 and EN 50600	Data-centre facilities, infrastructure, availability, security, efficiency and management concepts.	PHY, POW, THM, ENV, OPS and RES.	No clause-level equivalence claimed.
ANSI/TIA-942-C	Telecommunications, power, cooling, architecture, fire protection, safety and physical security.	PHY, POW, THM and NET.	No TIA certification claim.
Uptime Tier Topology	Site-infrastructure capacity components and distribution paths across Tier classifications.	POW, THM, PHY and OPS design evidence.	Tier is not converted to capability assurance.
ISA/IEC 62443	Requirements and processes for electronically secure IACS across lifecycle, operations and technology.	SEC, OPS and SUP.	Cybersecurity proof does not establish facility resilience.
NIST AI RMF	Voluntary AI risk-management and trustworthiness functions.	GOV, CAP, SEC and EVA linkage.	AIDCF is complementary, not a NIST profile.
ASHRAE data-centre resources	Cooling, energy, environmental and technology-space design/operation guidance.	THM and ENV evidence.	Engineering criteria remain source-owned.
Reliability and safety techniques	FMEA/FMECA, fault trees, event trees, bow ties and structured hazard analysis.	DEP, RES and engineering domains.	Competent method use required.
Assurance-case methods	Structured claim, argument and evidence patterns.	AIDCF assurance graph and reporting.	Formal GSN mapping remains future work.
W3C PROV-O and semantic standards	Provenance and graph representation.	Evidence lineage and machine exchange.	Semantic conformance does not prove truth.
OCP and vendor AI reference architectures	Open or validated implementation patterns for AI infrastructure.	Architecture evidence and interface patterns.	Reference design does not prove site operation.

36. Control-universe completeness and symmetry test

The 15/90/360 structure is an AIDCF-internal identity and cardinality baseline. Equal family and control counts are catalogue-design constraints supporting stable identity and machine processing. They are not evidence that all domains possess equal inherent complexity.

Test	Method	Acceptance
External requirement coverage	Map source requirements and concepts to domains/families/controls.	No material orphan without documented exclusion.
Incident coverage	Map representative AI-infrastructure incidents and failure scenarios.	Every material path maps or triggers extension.
Duplicate analysis	Compare objectives, conditions, evidence and effects.	Duplicates merged or differentiated explicitly.
Padding test	Challenge low-density families and generic control wording.	No control retained solely to satisfy cardinality.
Compression test	Challenge high-complexity domains for overloaded controls.	Split when one control has materially distinct evidence and ownership.
Pilot orphan test	Record findings that cannot be classified.	Unmapped material finding creates change proposal.

37. Scoring re-performance specification

AIDCF-010 states that raw and adjusted indicators, coverage, confidence, gate overlays and accountable decisions remain separate, and that scores do not create assurance. The whitepaper therefore defines a minimum re-performance record without inventing numeric weights not present in approved source evidence.

Field	Required value
Capability/profile	Stable ID, version and applicable control set.
Control input	Dimension results, applicability, assessor and reviewer.
Evidence overlay	Evidence-quality index, sufficiency and confidence.
Coverage	Applicable, numerically assessed, UNKNOWN, not assessed and approved N/A.
Formula version	Exact configuration and weight set.
Raw indicator	Deterministic output before evidence/gate effects.
Adjusted indicator	Separately labelled output after declared evidence rule.
Gate overlay	All applicable gate states and controlling reasons.
Recommendation	Engine output with reasons.
Decision	Human authority, review, limitations, validity and triggers.
Re-performance	Input hash, tool version, output hash and independent result.

Current evidence

The synthetic package contains a score snapshot and deterministic score re-performance output, but v3.0 does not reproduce an exact numeric calculation because the necessary source values were not available in the retrieved excerpts. Fabricating weights or scores would violate the framework's evidence doctrine.

38. Technical architecture viewpoints

Viewpoint	Mandatory elements	Review question
Service outcome	Capability, users, outcomes, tolerances and criticality.	What result must be protected?
Physical facility	Site, space, power, cooling, fluid, environment and physical routes.	What physical infrastructure enables the outcome?
AI technology	Compute, scheduler, storage, metadata, fabrics, gateways and management planes.	What configured technology enables the outcome?
Authority	Identities, privileges, automation, remote support and emergency control.	Who or what can change the state?
Dependency/fault domain	Shared sources, controls, routes, people, suppliers and recovery resources.	What can fail together?
Evidence	Telemetry, configuration, tests, documents, human evidence and independent evidence.	What proves each assertion?
Failure and recovery	Initiation, propagation, containment, recovery and residual path.	How is the outcome lost and restored?
AITG handoff	Shared identity, boundary interface, trust/authority path and federated decision.	Where do infrastructure and AI trust reasoning meet?

39. Full scenario-record example

Field	Synthetic example
Scenario ID	SCN-SYN-THM-001
Capability outcome	Distributed-training job completion and controlled thermal operation.
Initiating condition	Synthetic reduction in CDU flow.
Preconditions	Declared high-load training state and dependence on the affected loop.
Local failure mode	Reduced coolant flow at the synthetic cooling distribution unit.
Propagation	Temperature rise → accelerator throttling → collective delay → checkpoint exposure.
Affected states	Normal transitions to degraded; recovery path required.
Safeguards	Flow alarm, temperature telemetry, load reduction, isolation and recovery procedure.
Safeguard dependencies	Sensor integrity, management network, operator authority and alternate capacity.
Residual path	Common controller or insufficient alternate thermal capacity.
Evidence requirements	P&ID, heat-load model, flow/temperature telemetry, alarms, operator record and authorised test.
Gate effects	CG-06, CG-08, CG-09 and CG-10.
Safety boundary	No live reduction or switching without explicit competent authorisation.
Reassessment trigger	Thermal design, firmware, load, provider, incident or evidence change.

40. Information-classification and disclosure model

Class	Permitted content	Restricted examples
PUBLIC	Framework concepts, synthetic examples, public references and aggregate validation status.	No real site paths, credentials or exploitable weaknesses.
CONTROLLED	Detailed control mappings, reviewer comments and non-sensitive pilot summaries.	Distribution limited to authorised reviewers.
CONFIDENTIAL	Architecture, provider boundaries, evidence, findings and remediation.	No public deposit.
HIGHLY RESTRICTED	Privileged paths, physical routes, emergency procedures, vulnerabilities and sensitive telemetry.	Need-to-know access and separate secure repository.

- Public reports may redact detail but must not hide decision limitations, material gate FAIL/UNKNOWN or scope restrictions.
- Graph exports must filter restricted nodes, relationships and properties according to classification.
- Evidence references should use controlled identifiers rather than embedding secrets or sensitive records.
- Public benchmark cases must be synthetic or demonstrably authorised and anonymised.

41. Independent review board charter

Element	Requirement
Purpose	Challenge technical adequacy, evidence doctrine, claims, safety, scoring, semantics and usability.
Independence	Reviewers disclose conflicts, prior design involvement and commercial interest.
Competence	Collective electrical, mechanical, AI/HPC, storage/network, operations, OT security, assurance, measurement, legal/safety and semantic-web competence.
Method	Issue-based review using severity, evidence, recommendation and disposition.
Decision	Accept, accept with condition, reject, or defer each material disposition.
Transparency	Publish non-sensitive comments and dispositions with reviewer consent.
Authority	Review board advises; framework governance approves releases.
Stop rule	Material unresolved safety, evidence-integrity or claim issue prevents final release.

42. Public review instructions and issue taxonomy

A real public-review release must publish a review channel and dates in the repository or deposit metadata. Those details cannot be invented in this document and remain placeholders until selected by the author.

Issue type	Use
TECH	Technical validity, engineering or AI-infrastructure issue.
ASSUR	Evidence, assurance, gates, scoring or decision issue.
SEM	Ontology, graph, identity or machine-validation issue.
SAFE	Safety, legality, privacy or security-disclosure issue.
ED	Editorial, terminology or structure issue.
IMPL	Implementation, tooling or interoperability issue.
PILOT	Pilot design, feasibility or measurement issue.

Required comment field	Content
Comment ID	Stable identifier.
Section/object	Exact artifact version and location.
Issue	Concise statement.
Evidence	Source or reasoning.
Severity	Blocker, major, moderate or editorial.
Recommendation	Proposed change.
Disposition	Accept, partial, reject or defer with rationale.
Reviewer response	Acceptance or objection.
Closure evidence	Changed artifact, test or decision record.

43. Authorised pilot programme package

The user requested at least one authorised pilot. No signed AIDCF pilot charter, named sponsor, authorised site or completed pilot result was found in the available AIDCF sources. Therefore v3.0 includes the complete pilot-authorisation and execution package but does not falsely claim that a real pilot has been authorised or completed.

Pilot status
PILOT-001: NOT AUTHORISED / NOT EXECUTED. This is a controlled template and execution plan, not a pilot result.

Pilot field	Required content
Pilot ID	PILOT-001
Sponsor	Named accountable organisational sponsor.
Capability owner	Owner of the bounded AI Capability Instance.
Decision question	Exact intended decision and prohibited interpretation.
Scope	Physical, logical, organisational, provider and temporal boundary.
Operating states	Normal, maintenance, degraded, emergency, recovery and transition as applicable.
Profile	Control, gate, evidence and assurance-depth profile.
Authorisation	Signed scope, access, safety, testing and disclosure approval.
Team	Lead assessor, domain SMEs, evidence lead, independent reviewer and decision authority.
Data handling	Classification, retention, redaction and permitted publication.
Stop criteria	Unsafe condition, unresolved authority, source-integrity failure or scope breach.
Outputs	Charter, plan, evidence manifest, workpapers, gates, decision, report and lessons.

43.1 Pilot acceptance criteria

- Every assessed assertion has a stable identity and evidence reference.
- All applicable gates have completed workbooks and independent review.
- UNKNOWN, exclusions, conflicts and inaccessible scope remain visible.
- Coverage is calculated against the approved applicable profile.
- No test exceeds written authority or competent engineering governance.
- The issued decision is bounded by scope, states, period, evidence and intended users.
- Any public case material has explicit disclosure approval.

44. Pilot execution plan and measurement framework

Phase	Activity	Exit evidence
Authorise	Approve charter, roles, safety, access, data handling and publication.	Signed pilot authorisation.
Profile	Define outcomes, architecture, boundaries, states, controls and gates.	Approved pilot profile.
Plan	Methods, samples, tests, reviewers and stop rules.	Assessment plan.
Collect	Acquire evidence with provenance, integrity and restrictions.	Evidence manifest.
Assess	Execute control procedures and scenario analysis.	Reviewed workpapers.
Gate and score	Evaluate gates and reproduce score.	Gate register and score snapshot.
Decide	Independent review and accountable decision.	Signed bounded decision.
Report	Issue controlled reports and public summary if authorised.	Report package.
Learn	Record usability, disagreements, orphan findings and changes.	Pilot lessons and change proposals.

Measure	Definition
Scope agreement	Inter-rater consistency in capability and boundary definition.
Trace completeness	Required chain links present and reverse-queryable.
Evidence effort	Effort to collect, validate and review evidence.
Gate agreement	Inter-rater gate-state agreement and disagreement causes.
Control agreement	Inter-rater control-result agreement.
Orphan findings	Material findings without a suitable domain/control mapping.
Decision usefulness	Named user assessment of relevance and clarity.
Economic signal	Observed accelerator-hours, interruption, recovery or evidence-friction measures where authorised.

45. Pilot authorisation and result record

Approval/result	Status	Required evidence
Sponsor authorisation	PENDING	Signed charter
Capability-owner authorisation	PENDING	Scope and evidence-access approval
Safety/testing authorisation	PENDING	Rules of engagement and stop criteria
Data/privacy/security approval	PENDING	Classification and disclosure decision
Independent reviewer acceptance	PENDING	Review memorandum
Execution	NOT STARTED	Workpapers and logs
Assurance Decision	NOT ISSUED	Signed decision record
Public summary	NOT AUTHORISED	Publication approval

This record is intentionally incomplete. Completing it requires real organisational authority, named reviewers and evidence that cannot be fabricated by the whitepaper author or the drafting assistant.

46. AITG composed scenario example

Layer	Synthetic composed path
AIDCF infrastructure path	Cooling-control dependency degrades thermal capacity and affects accelerator operation.
Boundary handoff	The cooling-management service is referenced as a dependency and enforcement point for the AI capability.
AITG path	Authority over the management interface and remote-support relationship are examined as trust and exposure relationships.
Evidence reuse	Access logs and configuration records support separate AIDCF and AITG assertions with distinct relevance reviews.
Native findings	AIDCF reports infrastructure safeguard and recovery effects; AITG reports authority/exposure effects.
Federated decision	Both conclusions appear in one package without automatic aggregation.

47. Publication and archival release gate

Release item	Required before final v3.0
Table of contents and navigation	Generate and update fields in final PDF/DOCX.
Licence	Select whitepaper, dataset and software licences separately.
DOI	Reserve DOI and insert into final artefacts and metadata.
Independent review	Close or disclose all blocker/major issues.
Gate approval	Publish signed canonical gate registry.
Pilot	Complete at least one genuinely authorised pilot or retain validation-candidate status.
Accessibility	Alt text, tagged reading order, table headers, bookmarks and contrast.
Archival PDF	Produce stable PDF/PDF-A as appropriate with embedded fonts.
Manifest	Publish checksums, versions and related-output identifiers.
Review channel	Publish issue/contact channel and disposition policy.

Release rule
If independent gate approval and an authorised pilot remain incomplete, the document must retain VALIDATION CANDIDATE or PUBLIC REVIEW DRAFT status. It must not be presented as a final validated framework.

48. v3.0 conclusion

AIDCF v3.0 is a comprehensive validation candidate. It integrates the conceptual model, control universe, evidence and assessment doctrine, gate-controlled decision model, machine-executable traceability, reviewer dispositions, candidate gate registry, derivation-record package, security disclosure model, independent-review charter and pilot programme package.

The remaining work is evidential, not editorial: named independent reviewers must accept or challenge the dispositions; the critical gates must be ratified through documented governance; the derivation registers must be populated and reviewed; scoring must be re-performed from disclosed source values; and at least one real pilot must be authorised and executed. Until those records exist, v3.0 is not described as externally validated.

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